

Erin Zhang

+1 (647) 712-6722 | y3695zha@uwaterloo.ca | linkedin.com/in/erin-zhang | github.com/erin-zhang1

Personal Website: built-by-erin.vercel.app

SUMMARY

Expected University of Waterloo graduate with double majors in Mathematical Finance and Statistics and a Computing Minor, focused on **Python backend development**, **REST APIs**, internal tools, and correctness-critical financial data systems. Experience building DjangoFastAPI workflows, SQL-backed pipelines, and pytest-covered financial calculation systems, with a strong focus on correctness, reliability, and reproducible backend workflows.

EDUCATION

University of Waterloo

Expected Dec 2026

Double Major in Mathematical Finance and Statistics, with Computing minor (Computer Science) Waterloo, Canada

- **President's Scholarship**
- Relevant Coursework: Algorithms & Data Structures, Database Systems, Applications Software Engineering, Data-Intensive Distributed Analytics
- Co-op Performance: Earned **Excellent or Outstanding** employer evaluations across all 6 co-op work terms.

TECHNICAL SKILLS

Backend: Python, FastAPI, Django, REST APIs, PostgreSQL, MySQL, SQLAlchemy, Alembic, JWT, request validation

Data & Automation: pandas, NumPy, ETL pipelines, FTP/SFTP workflows, DBF/CSV processing

Testing & CI/CD: pytest, unit testing, integration testing, regression testing, data validation, GitHub Actions

Infrastructure: Git/GitHub, Linux, Docker, Prometheus, Bash

EXPERIENCE

Actuarial Software Developer (Python & R)

Jan 2026 – Present (Expected Aug 2026)

Atella Consulting

North Carolina, United States

- Owned end-to-end development of a **Python/R backend automation system** powering **10 recurring actuarial reporting workflows**, reducing daily calculation runtime from **5+ hours** of manual Excel processing to **~3 seconds**.
- Built backend features for a **Django-based internal web application** and implemented **RESTful API endpoints** with request validation, database query optimization, and integration with downstream reporting jobs.
- Expanded **pytest unit and regression test coverage to 94%** for core actuarial calculation modules, catching regressions in policy mapping, rate selection, interest accrual, and reporting outputs.
- Designed snapshot-based reporting infrastructure with automated **FTP/SFTP ingestion**, DBF/CSV processing, timestamped outputs, and structured run metadata for daily and month-end traceability.

Data Reporting Analyst (SQL & Python)

Sep 2024 – Apr 2025

Kenna Communications

Mississauga, Canada

- Developed automated reporting pipelines in **Python and SQL** that consolidated data from three sources, eliminated **15 hours** of manual work per week, and delivered recurring operational reports.

Data Research Analyst (R & Linux)

Jan 2024 – Apr 2024

Chinese Academy of Sciences - Lanshi Institute of AI in Medicine

Remote

- Developed automated QA pipelines for **10TB** of genomic sequencing data using R and Linux, improving reproducibility and ensuring **98%** downstream analysis accuracy.

Quantitative Analyst Intern (Python)

Sep 2023 – Dec 2023

AssetPro

Shanghai, China

- Built a Python backtesting framework for factor-based trading signals and parallelized market-data pipelines, reducing daily processing time from **2 hours** to under **15 minutes**.

PROJECTS

TaskFlow API: JWT-Auth Backend Service

FastAPI, PostgreSQL, Docker, pytest, SQLAlchemy, Alembic, JWT

- Built a **RESTful task-management API** with FastAPI, including user registration, **JWT authentication**, role-based access control, CRUD endpoints, and OpenAPI documentation.
- Modeled data with **SQLAlchemy** and Alembic migrations, containerized the API/PostgreSQL stack with **Docker Compose**, and added **GitHub Actions CI** to run pytest coverage checks across auth, validation, error handling, and rollback tests.